

Matrix Computations

Lecture 04: More on SVD

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September 12, 2026



Outline

- 1 SVD and eigenvalue decomposition
- 2 Low-rank approximation
- 3 Perturbation bounds for singular values
- 4 Nuclear norm

Part 1. SVD and eigenvalue decomposition

SVD and the rank of a matrix

- The rank of A is the number of nonzero singular values of A

$$\text{rank}(A) = \text{rank}(U\Sigma V^H) = \text{rank}(\Sigma)$$

- Let $A \in R^{m \times n}$, and $m \geq n$, if $\text{rank}(A) < n$, A is said to be rank-deficient
- In practice, A can be close to rank-deficient, or near-rank-deficient: even though all the n singular values are positive, some of them are relatively small compared to the larger ones
- Elbow-plot of singular values

Four fundamental subspaces associated with A exposed by SVD

Let SVD of $A \in C^{m \times n}$ be $A = U\Sigma V^H$

- Let $r = \text{rank}(A)$, and $U = [u_1, \dots, u_m]$ and $V = [v_1, \dots, v_n]$, then

$$\mathcal{C}(A) = \text{span}\{u_1, \dots, u_r\}, \quad \mathcal{N}(A) = \text{span}\{v_{r+1}, \dots, v_n\}$$

$$\mathcal{C}(A^H) = \text{span}\{v_1, \dots, v_r\}, \quad \mathcal{N}(A^H) = \text{span}\{u_{r+1}, \dots, u_m\}$$

- It's also easy to see

$$\mathcal{C}(A)^\perp = \mathcal{N}(A^H), \quad \mathcal{C}(A^H)^\perp = \mathcal{N}(A)$$

and the above implies

$$C^m = \mathcal{C}(A) \oplus \mathcal{N}(A^H), \quad C^n = \mathcal{C}(A^H) \oplus \mathcal{N}(A)$$

where $\oplus$ indicates **direct sum** of two subspaces

SVD and eigenvalue decomposition

- Let SVD of A be $A = U\Sigma V^H$, then

$$A^T A = V(\Sigma^T \Sigma)V^T, \quad AA^T = U(\Sigma \Sigma^T)U^T$$

are the eigenvalue decomposition of $A^T A$ and AA^T , respectively

- Let A be Hermitian, and $A = Q\Lambda Q^H$ the eigenvalue decomposition of A . Let $\lambda_j = |\lambda_j| \exp(i\theta_j)$, $i = \sqrt{-1}$. Then the SVD of A is

$$A = [Q \text{diag}(\exp(i\theta_1), \dots, \exp(i\theta_m))] \underbrace{\text{diag}(|\lambda_1|, \dots, |\lambda_m|)}_{\Sigma} Q^H$$

- $|\det(A)| = \prod_{i=1}^m \sigma_i$

SVD and eigenvalue decomposition

- Let

$$H = \begin{bmatrix} 0 & A^T \\ A & 0 \end{bmatrix}$$

Then the $2n$ eigenvalues of H are $\pm\sigma_i, i = 1, \dots, n$, and the corresponding unit eigenvectors are

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} v_i \\ \pm u_i \end{bmatrix}$$

Part 2. Low-rank approximation

Rank-one matrices and rank- r matrices

- Let $A = uv^H$, $u \in C^m$, $v \in C^n$, and u and v are nonzero. The SVD of A is

$$A = \frac{u}{\|u\|_2} (\|u\|_2 \|v\|_2) \frac{v^H}{\|v\|_2} = (\|u\|_2 \|v\|_2) \frac{u}{\|u\|_2} \frac{v^H}{\|v\|_2}$$

This also shows $\|A\|_2 = \|A\|_F = \|u\|_2 \|v\|_2$

- More generally let SVD of A be $A = U\Sigma V^H$, and $r = \text{rank}(A)$, and $U = [u_1, \dots, u_m]$ and $V = [v_1, \dots, v_n]$, then

$$A = U_r \Sigma_r V_r^T = \sum_{i=1}^r \sigma_i u_i v_i^H$$

the sum of r rank-one matrices, where

$$U_r = [u_1, \dots, u_r], \quad V_r = [v_1, \dots, v_r], \quad \Sigma_r = \text{diag}(\sigma_1, \dots, \sigma_r)$$

Rank- k approximations

- For $k \leq r$, let

$$A_k \equiv \sum_{i=1}^k \sigma_i u_i v_i^H$$

A_k is a rank- k matrix, it turns out to have an optimality property

$$\|A - A_k\|_2 = \min_{\text{rank}(B) \leq k} \|A - B\|_2 = \sigma_{k+1}, \quad \sigma_{r+1} = 0$$

- **Proof.** Since $\text{rank}(B) \leq k$, $\dim(\mathcal{N}(B)) \geq n - k$.
- Now the dimension of $S = \text{span}\{v_1, \dots, v_{k+1}\}$ is $k + 1$
- There are nonzero vectors in the intersection $\mathcal{N}(B) \cap S$ (proof in class)

Best rank- k approximation

- **Proof** (continued). Let $v \in \mathcal{N}(B) \cap S$ and $\|v\|_2 = 1$. Then

$$Bv = 0, \quad v = \sum_{i=1}^{k+1} \alpha_i v_i, \quad \sum_{i=1}^{k+1} |\alpha_i|^2 = 1$$

- Now

$$\begin{aligned} \|(A - B)v\|_2^2 &= \|Av\|_2^2 = \left\| \sum_{i=1}^{k+1} \alpha_i \sigma_i u_i \right\|_2^2 \\ &= \sum_{i=1}^{k+1} |\alpha_i|^2 \sigma_i^2 \geq \sigma_{k+1}^2 \sum_{i=1}^{k+1} |\alpha_i|^2 = \sigma_{k+1}^2 \end{aligned}$$

which implies $\|A - B\|_2 \geq \sigma_{k+1}$. It's also easy to verify $\|A - A_k\|_2 = \sigma_{k+1}$

Part 3. Perturbation bounds for singular values

Characterization and perturbation bounds of singular values

- Let the singular values of $A \in R^{m \times n}$ be $\sigma_1 \geq \dots \geq \sigma_n$. Let S denote a subspace of R^n , then

$$\sigma_i = \max_{\dim(S)=i} \min_{x \in S, \|x\|_2=1} \|Ax\|_2$$

- Proof.** let SVD of A be $A = U\Sigma V^H$. First, let $S_0 = \text{span}\{v_1, \dots, v_i\}$. It can be shown

$$\min_{x \in S_0, \|x\|_2=1} \|Ax\|_2 = \sigma_i$$

Therefore,

$$\max_{\dim(S)=i} \min_{x \in S, \|x\|_2=1} \|Ax\|_2 \geq \sigma_i$$

Characterization and perturbation bounds of singular values

- Now for an arbitrary $\dim(S) = i$, it is easy to see the intersection

$$S \cap \text{span}\{v_i, \dots, v_n\}$$

contains nonzero vectors, and let v with $\|v\|_2 = 1$ be one of them, and

$$v = \sum_{k=i}^n \alpha_k v_k, \quad \sum_{k=i}^n |\alpha_k|^2 = 1$$

and

$$\|Ax\|_2^2 = \sum_{k=i}^n \sigma_k^2 |\alpha_k|^2 \leq \sigma_i^2 \implies \min_{x \in S, \|x\|_2=1} \|Ax\|_2 \leq \sigma_i$$

Therefore,

$$\max_{\dim(S)=i} \min_{x \in S, \|x\|_2=1} \|Ax\|_2 \leq \sigma_i$$

Characterization and perturbation bounds of singular values

- Perturbation bounds for singular values: $A, B \in R^{m \times n}$, and their singular values are

$$\sigma_1(A) \geq \cdots \geq \sigma_n(A), \quad \sigma_1(B) \geq \cdots \geq \sigma_n(B)$$

Then

$$\max_i |\sigma_i(A) - \sigma_i(B)| \leq \|A - B\|_2,$$

which can be proved using the max-min characterization of the singular values, and

$$\sum_{i=1}^n |\sigma_i(A) - \sigma_i(B)|^2 \leq \|A - B\|_F^2$$

which can be proved by Wieland-Hoffmann theorem.

Part 4. Nuclear norm

Singular value vector and Schatten p -norm

- Let the singular values of A be

$$\sigma_1 \geq \cdots \geq \sigma_{\min\{m,n\}}$$

- The singular value vector of A is

$$\sigma = [\sigma_1, \dots, \sigma_{\min\{m,n\}}]$$

- $\|A\|_2 = \|\sigma\|_\infty$, $\|A\|_F = \|\sigma\|_2$, and more generally Schatten p -norm of matrix A is defined to be

$$\left(\sigma_1^p + \cdots + \sigma_{\min\{m,n\}}^p\right)^{1/p} = \|\sigma\|_p$$

and letting $p \rightarrow 0$, we obtain

$$\text{rank}(A) = \|\sigma\|_0$$

Characterization of nuclear norm

- Schatten 1-norm is also called nuclear norm or trace norm

$$\|A\|_* = \sigma_1 + \cdots + \sigma_{\min\{m,n\}}$$

- **Proposition.**

$$\|A\|_* = \min_{A=XY^H} \|X\|_F \|Y\|_F = \min_{A=XY^H} \frac{1}{2} (\|X\|_F^2 + \|Y\|_F^2)$$

- **Proof.** Let SVD of A be $A = U\Sigma V^H$. For $A = XY^H$

$$\Sigma = U^H A V = (U^H X)(V^H Y)^H$$

Characterization of nuclear norm

- **Proof** (continued). Let $U^H X = [x_1, \dots, x_m]^H$ and $V^H Y = [y_1, \dots, y_n]^H$, then

$$\sigma_i = |x_i^H y_i|, \quad i = 1, \dots, p$$

- Use Chauchy-Schwarz

$$\sigma_i = |x_i^H y_i| \leq \|x_i\|_2 \|y_i\|_2$$

- Now assum $m \geq n$

$$\|A\|_* = \sigma_1 + \dots + \sigma_n \leq \|x_1\|_2 \|y_1\|_2 + \dots + \|x_n\|_2 \|y_n\|_2$$

Characterization of nuclear norm

- **Proof** (continued). Use Chauchy-Schwarz one more time

$$\begin{aligned}\|A\|_* &\leq \left(\sum_{i=1}^n \|x_i\|_2^2 \right)^{1/2} \left(\sum_{i=1}^n \|y_i\|_2^2 \right)^{1/2} \\ &\leq \|U^H X\|_F \|V^H Y\|_F = \|X\|_F \|Y\|_F\end{aligned}$$

because U and V are unitary

- Now let $X = U\Sigma^{1/2}$, and $Y = \Sigma^{1/2}V^H$, then

$$A = XY, \quad \|X\|_F = \|A\|_*^{1/2}, \quad \|Y\|_F = \|A\|_*^{1/2}$$